

MATH 471 COMPUTING PROJECT I

September 1, 2026 The object of this project is to apply Aitken's Δ^2 method to accelerate the convergence of a sequence.

The sequence

$$\begin{cases} p_{n+1} &= \sqrt{3 + p_n}, \quad n = 0, 1, 2, \dots \\ p_0 &= 1, \end{cases}$$

can be shown to converge to $p = \frac{1 + \sqrt{13}}{2}$.

Part I: Theory/Analysis

- (i) Show that the sequence $\{p_n\}$ satisfies $1 \leq p_n \leq 3$
- (ii) Show that $\{p_n\}$ is increasing. Conclude that it converges to a limit $p \leq 3$.
- (iii) Show that the limit is indeed given by $p = \frac{1 + \sqrt{13}}{2}$.
- (iv) Compare $p_{n+1} - p$ to $p_n - p$ and evaluate the asymptotic constant λ

Part II: Numerics

- (1) Generate the sequence $p_0, \dots, p_{10}$
- (2) Generate the sequence $\left| \frac{p_{n+1} - p}{p_n - p} \right|$, $n = 0, \dots, 9$
- (3) Generate the sequence $\hat{p}_n = p_n - \frac{(p_{n+1} - p_n)^2}{p_{n+2} - 2p_{n+1} + p_n}$, $n = 0, \dots, 8$ using Aitken's formula.
- (4) Generate the sequence $\left| \frac{\hat{p}_{n+1} - p}{\hat{p}_n - p} \right|$, $n = 0, \dots, 7$

Use double precision in your calculations.

Place the computed numbers in the indicated columns as in the table.

Show 16 digits in columns 2 and 4 but 6 digits in columns 3, 5 and 6.

Leave the dashed cells empty.

n	p_n	$\frac{p_{n+1} - p}{p_n - p}$	$\hat{p}_n$	$\frac{\hat{p}_{n+1} - p}{\hat{p}_n - p}$	$\frac{\hat{p}_n - p}{p_n - p}$
0					
1					
2					
3					
4					
5					
6					
7					
8				_____	
9			_____	_____	_____
10		_____	_____	_____	_____

Answer the following questions

- (i) What is the value of the asymptotic convergence rate λ for each of the two sequences p_n and $\hat{p}_n$?
- (ii) Do the rates indicate that the sequence $\{\hat{p}_n\}$ converges faster than p_n ?
- (iii) Would you be correct to conclude that $\{\hat{p}_n\}$ converges superlinearly?